

Internship Progress Report

Chronic Disease Burden & Social Determinants Dashboard

Date: March 30, 2026
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Organization: Global AI
Period: Week 2 — Part-Time Internship

Work Completed

This week was a lot more demanding than the first. Last week was mostly reading and understanding the data; this week I actually had to do things with it, and that turned out to be much harder than I expected. The plan was to finish cleaning, get the datasets merged, and start the correlation analysis. All of that is done, but it took basically the whole week.

Data Cleaning

I started by dealing with the issues flagged during the EDA. The two columns that were completely null in every file were dropped, along with the 66 rows in the 2025 release that had missing health values. The 2020 PLACES file was left out as planned since there was no way to match it to the SVI data without a FIPS code.

I also applied zero-padding to the county identifiers in both datasets. The actual code change was one line, but I spent a while checking that it had worked correctly across all files before moving on.

Reshaping and Merging

The pivot step took way longer than I thought it would. Each PLACES file has a slightly different set of health measures depending on when it was released, so I could not just pivot all the files at once. I had to handle each one separately, make sure the column names lined up, and then stack them into a single panel. That panel ended up being 15,720 rows across five release years (2021 to 2025) and 50 columns.

Joining it to the SVI data had its own complications, which I describe more in Section 2. Once the merge was done and I had removed counties missing the key outcome or vulnerability fields, the final dataset came out to **15,341 rows** and **59 columns**, saved to `processed_data/panel_county_health.csv`.

Correlation Analysis

Once I had a clean merged dataset I moved on to the first analytical task, which was looking at how social determinants of health relate to county-level diabetes prevalence and hypertension rates. I computed Pearson correlations between seven SDOH factors and both outcomes for each data year from 2019 to 2023, then built a 2023 snapshot heatmap covering all available SDOH and SVI factors. Four figures were saved to the `results/` folder:

- `fig1_sdohtrend.png`: five-year correlation trends for seven SDOH factors against both outcomes.
- `fig2_2019_vs_2023.png`: grouped bar chart comparing correlations at the 2019 and 2023 endpoints.
- `fig3_heatmap_2023.png`: full cross-sectional heatmap of all SDOH and SVI factors for 2023.

- `fig4_top_correlates_2023.png`: ranked bar chart of the strongest correlates of diabetes and hypertension in 2023.

I also built an interactive choropleth map showing diabetes prevalence by county across all five data years, with a year slider and a toggleable SVI overlay. It is saved as a self-contained HTML file at `fig5-diabetes-svi-map.html`.

Difficulties Encountered

Honestly, this was a really tough week. A lot of things that seemed simple going in were not simple at all once I started coding them.

- **Pivoting long to wide format:** Because each PLACES file has a different set of measures depending on the release year, I could not run a single pivot across all files at once. I had to pivot each one separately, make sure the column names matched up, and carefully verify that nothing had shifted when I stacked them. It was the kind of task where it is hard to tell if it worked correctly until something downstream breaks.
- **Matching SVI snapshots to PLACES years:** The SVI is only published in two snapshots (2020 and 2022), but the PLACES panel covers five release years. I had to read through the methodology documentation to figure out which SVI snapshot to attach to which PLACES year, and ended up joining 2021–2022 PLACES to SVI 2020 and 2023–2025 to SVI 2022. It felt like a reasonable call, but it is the kind of decision that could quietly break things if it is wrong.
- **Row loss after merging:** After the join, 365 rows had disappeared. They were counties present in PLACES but missing values for `DIABETES`, `BPHIGH`, or `RPL_THEMES`. I spent a while confirming that this was expected data sparsity and not a sign that the merge itself had gone wrong.
- **Setting up the correlation analysis:** Getting the correlation framework structured so it worked consistently across years, outcomes, and feature groups took more planning than I expected. Most of the time was not spent computing correlations but figuring out how to organise the code so it would not fall apart when I needed to extend it later.
- **Building the interactive map:** I had not used `folium` before, so this involved a lot of reading documentation and trial and error. Getting the FIPS codes to align between the health data and the GeoJSON boundary file, building separate feature layers per year, and wiring up the slider and SVI overlay all had to work together. 278 counties ended up skipped because their geometry was missing from the boundary file.

Planned Work

Next week I want to go through the correlation results more carefully and pull out the clearest patterns, then write a short summary to go alongside the figures. After that I will start on county-level segmentation, which means grouping counties into risk tiers based on their chronic disease burden and social vulnerability profiles. I think that part will also take longer than it looks on paper.